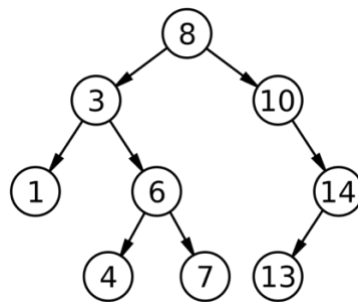


Midterm Exam – Sample Question

1. Perform insert and delete operations in a Linked List. Illustrate with examples. [4]
2. A hospital uses a queue to manage patient appointments. [6]
 - a) Illustrate how patients are inserted and removed in a linear queue.
 - b) Explain what happens when the queue becomes full but has empty slots at the front (illustrate with an example).
 - c) How does a circular queue solve this problem?
3. Consider the given binary tree and answer the following questions: [5]
 - a) Perform inorder, preorder, and postorder traversals and show the result.
 - b) If one node is deleted, explain which node will replace it in a complete binary tree.



4. You are given the following sequence of numbers to insert into a BST: [5]

50, 30, 70, 20, 40, 60, 80

- a) Construct the BST.
- b) Show the steps to search for the key 40.
- c) If node 30 is deleted, which node will replace it? Justify your answer.

Good Luck!!!